Fieldwork in the South African Savanna
Tiffany Taketa

Wading through fields of grass with blue skies ahead, soft sand and gravel crunching underfoot, carefully edging around waterholes with agitated hippos encroaching while we quickly hammer a cage stand into the ground, slash grass, record data, set up a camera trap, and coordinate where to put the next one…

Walking in the field at a camera trap site.

Part of the course involved field data collection and analysis using three different experimental methods: camera traps or trail cameras, vegetation surveys, and photovoice. During our time in South Africa, we designed basic experiments using camera traps and tested our hypotheses, as well as installed and maintained existing camera traps to assist Derek Fedak, a PhD candidate at Colorado State University, in his research on seasonal animal occupancy for the NASA Ecological Forecasting project.

For context, from the 1930s to the 1960s, artificial waterholes were installed in Kruger National Park (KNP) to abate the effects of a drought throughout the park and increase wildlife presence for tourism. This was done without the knowledge that a global drought caused by El Nino and La Nina effects was occurring. In the meantime, the increased number of waterholes caused problems such as overgrazing and led to the decline of certain megaherbivore populations. Thus, KNP has closed many of their artificial waterholes. However, Associated Private Nature Reserves (APNR), which are accessible to wildlife in the park, have not since certain species that are generally near waterholes can affect tourism. This has led some species, especially large numbers of elephants, to leave Kruger and move into APNRs, trampling shrubs and causing destruction to trees. With camera traps in both KNP and APNRs as well as data collected throughout the year, Derek's research investigates the effects that conservation management

decisions, such as installation of artificial waterholes, and climate change in the future could have on animal occupancy and movement.

Each camera, installed at different locations, gradients, and proximities to waterpoints, takes a burst of 3 photos with a 30 second quiet period between each photo, when triggered by movement (from an animal or unfortunately grass as well). After the memory cards were collected from camera traps, images were downloaded and organized onto a hard drive by members of our program, Haleigh and Hannah, to be analyzed later, since there are often thousands of photos per camera. Depending on the site, a wide variety of animals could be seen, from the most common species in the park to some of the rarest megafauna.

Maddy setting up a camera before installing it in a cage stand (on bottom left) and Ella recording site and camera information.

However, I think it's important to note that it's not just the technical knowledge we gained from this experience, but these hands-on experiences, in addition to firsthand encounters with wildlife and the opportunity to do a homestay in a village living near the reserve, provided a comprehensive understanding of the conservation operations carried out in one of Africa's largest game reserves, as well as the complex and ever-changing context surrounding the park and the often challenging to understand rhetoric around conservation in Africa itself, from a North American perspective. From the complex social dynamics created by the political history of the region and the former apartheid regime, the unique culture of communities surrounding the park, and limits that load-shedding imposed on our daily energy consumption and technology use due to the country's energy crisis – scientific research and conservation in KNP is much more than just wildlife and ecology. And I think the camera trap research on artificial waterholes, whose installation and closure is a result of the transition to a more decolonized approach to conservation, clearly illustrates the multi-dimensionality of conservation, especially in South Africa, and how ecosystems are handling conservation's changing philosophies.

As such, I think my understanding of the topics we discussed in this course would have been vastly different if I had taken a lecture-based course or visited South Africa's nearly 2 million ha Kruger National Park as an average tourist passing through. This is an experience I will remember forever, but not just because of the field and technical experience I gained, and not just because of everything we learned on this trip, but also because of the people I got to know on the program. And for all these reasons, I feel so grateful to have had this opportunity.